



Computer Vision for Autonomous Vehicles

Lecture 1: Introduction

*Instructor: Golnaz Habibi, PhD
University of Oklahoma
Fall 2025*



Welcome to CS4733/5733: Computer Vision for Autonomous Vehicle 2025!

- **About the Instructor:**

- Golnaz Habibi
- Assistant Professor, School of Computer Science
- Other Affiliation: DSAI, AME, ECE, ECE (University of Washington)
- Background: PhD in Computer Science, BSc/MSc Electrical Eng/ Control Systems
- Co-founder of VNAV Open source at MIT (Fall 2018)

- **Research interests:** Robotics, Computer Vision, Autonomous Navigation, Multi- Agent Systems, control systems, Machine Learning

- **Office Hours:**

Tu/Th 12-13, REPF B4 (robotics/mini-city lab)

Email: golnaz@ou.edu



Welcome to Computer Vision for Autonomous Vehicle 2025!

■ History of the Course:

- Originally called: Visual Navigation for Autonomous Vehicles (VNAV)
- Founded in Fall 2018 as a graduate level course at MIT
- The course became available as opensource since 2022
- All code bases became available since 2023
- Similar courses *has been offered in other universities around the world: TUM, University of Michigan and OU*
- *Previously offered at OU:*
 - CS 5970-003: Visual Navigation for Autonomous Vehicles, has been taught since Fall 2022

■ Related Article:

- Carlone. L, Khosoussi. K, Tzoumas. V, Habibi. G, Ryll. M, Talak. R, Shi. J, and Antonante. P, "Visual Navigation for Autonomous Vehicles: An Open-source Hands-on Robotics Course at MIT." *ISEC 2022*
- Habibi. G, Billerbeck. S, Vargas. D, Perez. D, McCulley, Evan, "Development and Implementation of a Hands-On Robotic Course on Computer Vision for Autonomous Vehicles", *ISEC 2025*

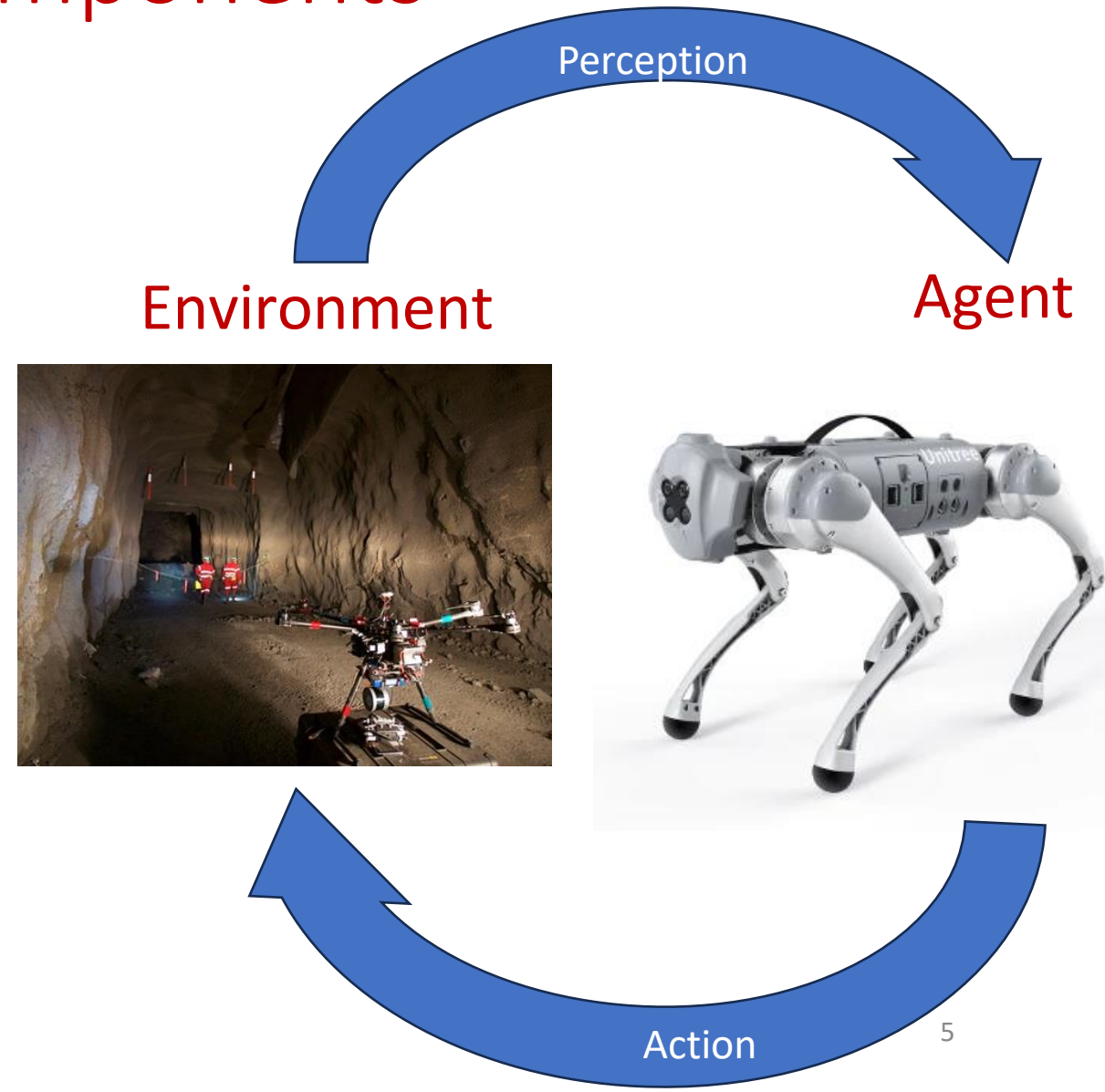
What is Autonomous navigation?

- *Autonomous Navigation is the problem of navigating an agent, e.g. a mobile robot, in an environment.*
- *The goal is to move from point **A** to point **B** by applying a sequence of actions.*

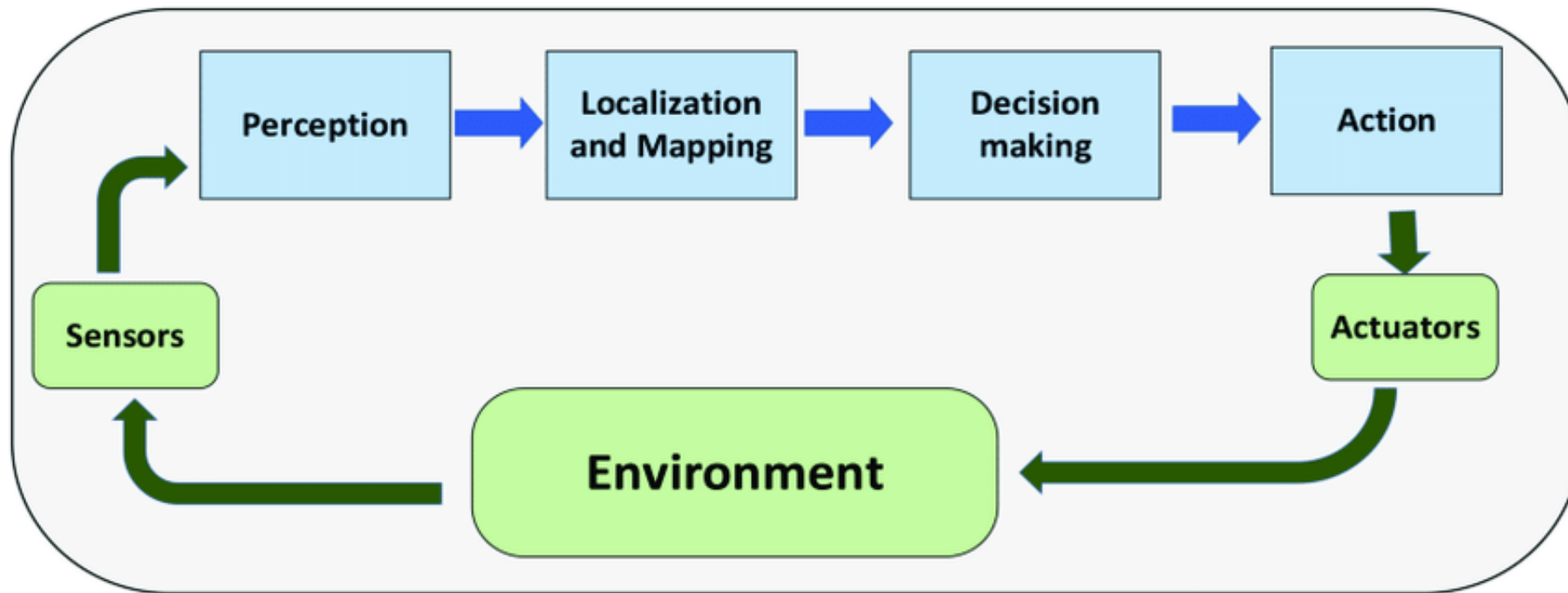
Kulhánek J, Derner E, De Bruin T, Babuška R. Vision-based navigation using deep reinforcement learning. In 2019 European Conference on Mobile Robots (ECMR) 2019 Sep 4 (pp. 1-8). IEEE.

Autonomous Vehicles Components

- **Perception:** understand the surroundings
 - Where am I? Localization
 - What does Environment look like
 - mapping
 - Object detection and tracking
- **Planning:** How can I reach a goal?
- **Control:** What is my action to follow the planned path
- Sometimes localization and mapping is considered separate from perception

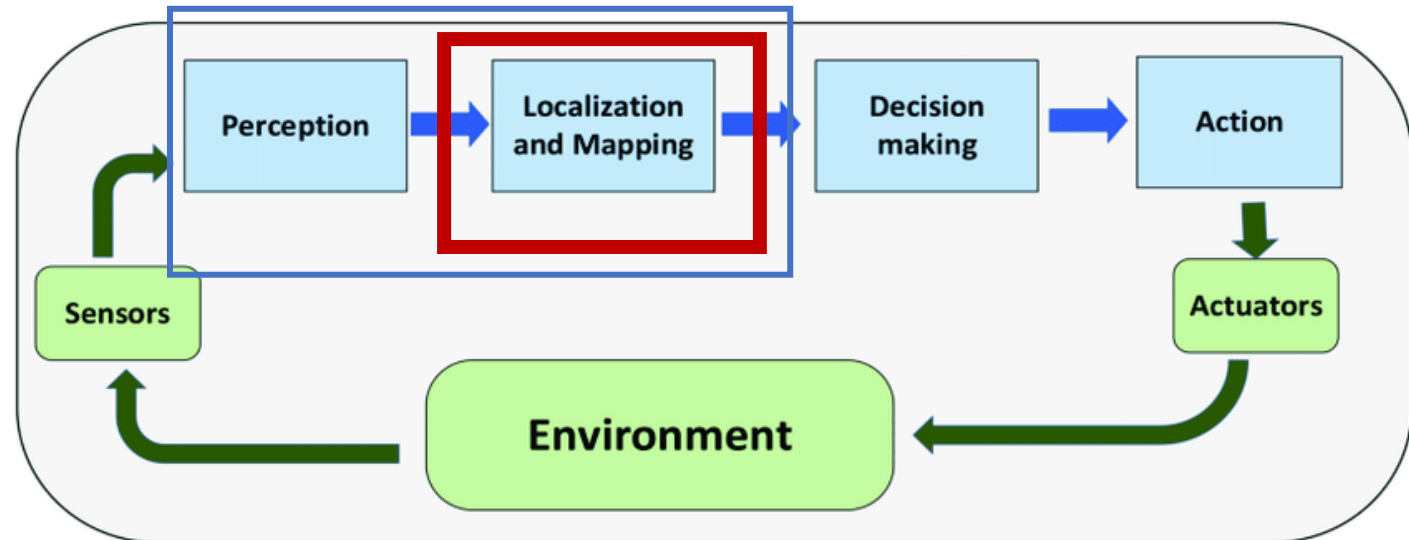


Autonomous Vehicles Components



What is this course about?

- Vision-based autonomous navigation
- This course covers all components of perception, planning, and control, however:
- The focus is to understand the world in 3D (i.e., perception module)
 - Mapping
 - Localization
 - Object detection and tracking



Robot Perception

real world

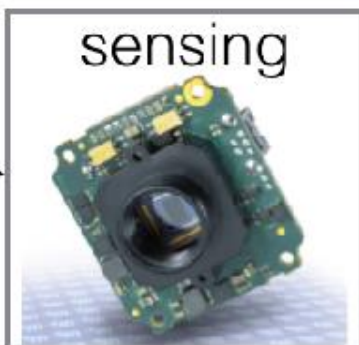


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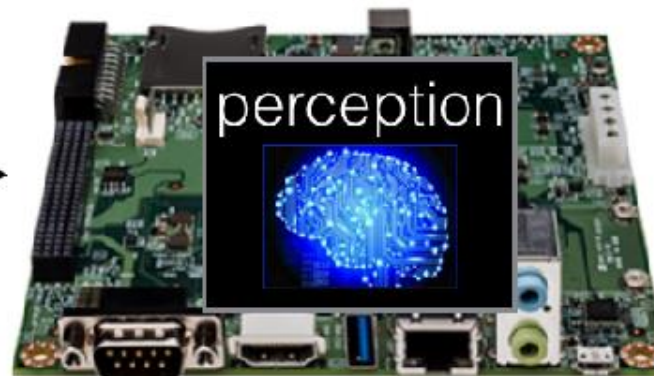
world model / representation



sensing



perception



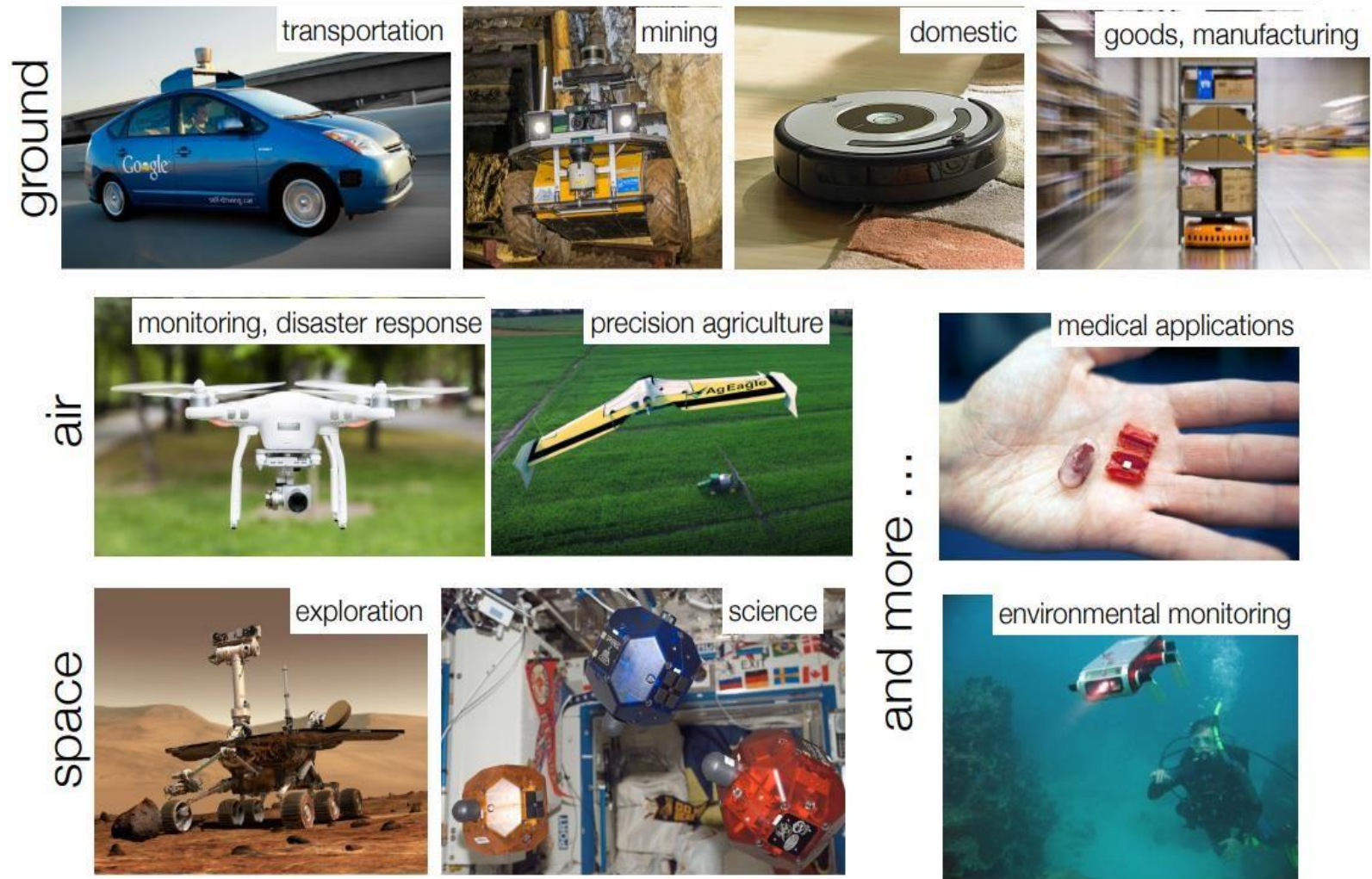
What is **Visual** navigation?

- ***Visual** Autonomous Navigation is the problem of navigating an agent, e.g. a mobile robot, in an environment using **camera** only*
- *The goal is to move from point **A** to point **B** by applying a sequence of actions, based on **the camera images only**.*

Kulhánek J, Derner E, De Bruin T, Babuška R. Vision-based navigation using deep reinforcement learning. In 2019 European Conference on Mobile Robots (ECMR) 2019 Sep 4 (pp. 1-8). IEEE.

Why Visual Navigation?

- Faster
- Less data/less process
- Less weight
- Less expensive
- More Accessible



Visual Autonomous Navigation: Legged Robot



Visual Autonomous Navigation: Self-driving cars



Visual Autonomous Navigation: drones



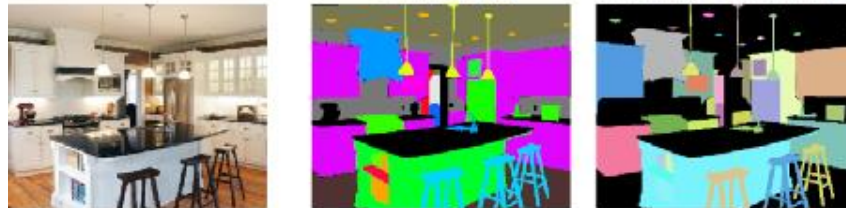
Why Perception Matters? Essential component of Safe Navigation



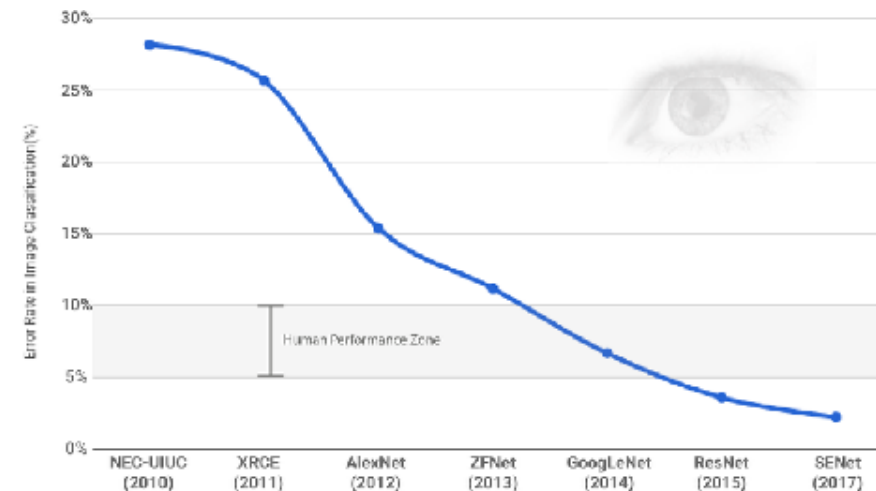
2.1. COCO Detection Challenge



3. Places Challenges



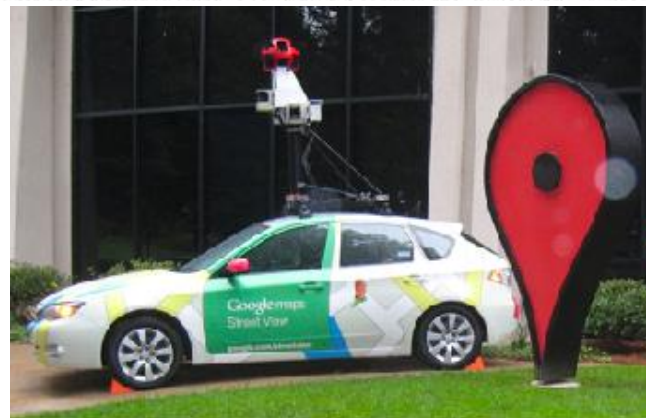
Things are starting to work!



Neural Network Architecture

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Google
Street
View



Oculus
Rift
Goggles



Perception success.. and its failures



“Google employs a small army of human operators to manually check and correct the maps”

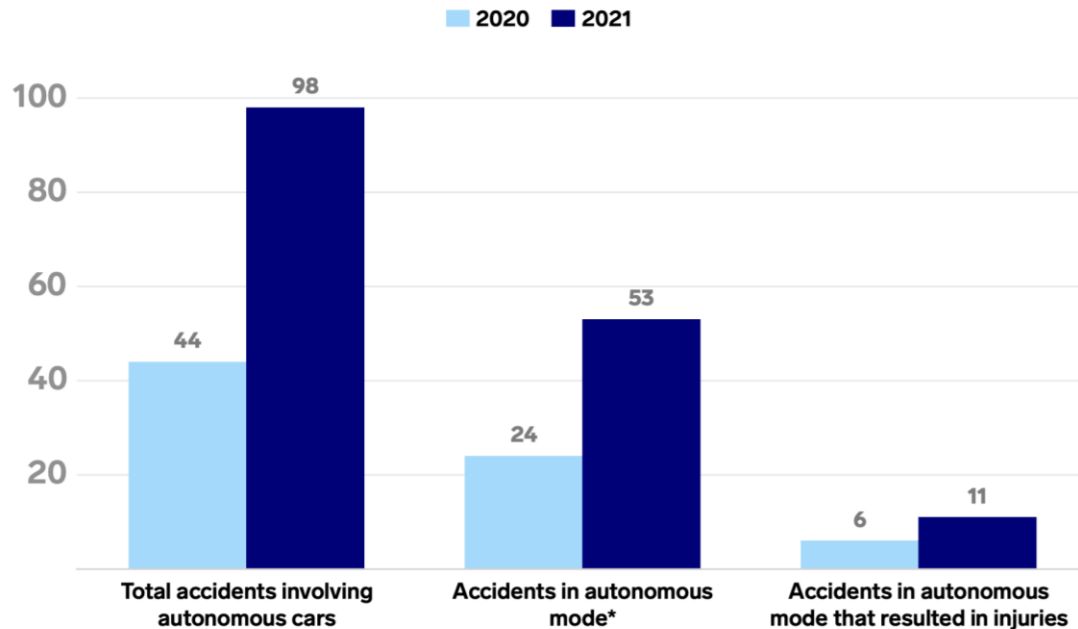
[Wired]

Slide credit: MIT VNAV 2023

Camouflage graffiti and art stickers cause a neural network to misclassify stop signs as speed limit 45 signs or yield signs. IMAGES: EVTIMOV ET AL

Robust Perception is crucial in Safety Critical Applications

Autonomous car collisions in California

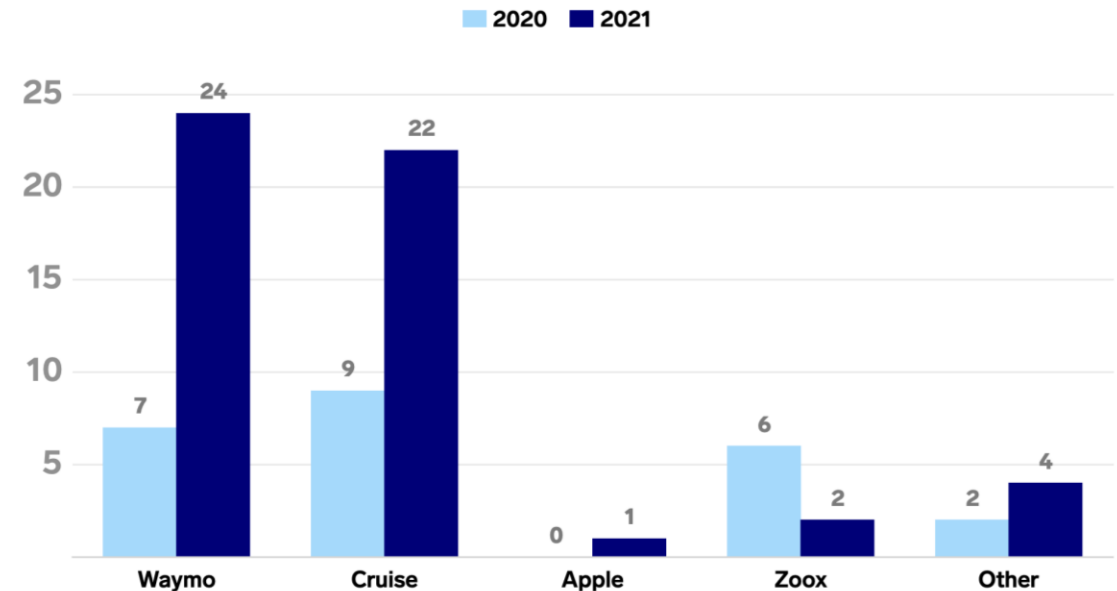


Note: *Insider describes autonomous mode as including cars reported to be in autonomous mode at time of accident, as well as those that had been in autonomous mode but switched to manual mode seconds before accident.

Source: California DMV

INSIDER

Car accidents in autonomous mode* in California, by company



Note: *Insider describes autonomous mode as including cars reported to be in autonomous mode at time of accident, as well as those that had been in autonomous mode but switched to manual mode seconds before accident.

Source: California DMV

INSIDER

Welcome to CS4733/CS5733!

This course ...

- Spans foundational principles of mathematics and the practical implementation of state-of-the-art computer vision for autonomous vehicles:
 - Self-driving cars
 - Delivery robots
 - Autonomous drones
- Intersection of Robotics and Computer Vision
- Hands-on robotic experience in 1/10th scale minicity
- Covers topics in:
 - Autonomous navigation, covering motion planning, computer vision, localization, and mapping

Course Objective I: Learn theory behind robot perception

- Learn the Learn/develop/exercise **theoretical tools to understand** state of the art in robot perception
- Focus: 2D computer vision (images), localization, mapping, 3D object pose estimation, and semantic understanding for visual navigation

Examples:

- Self-driving cars
- Drones
- Service robots
- Virtual and augmented reality



Course Objective II: Robotic Experience, implementation and Practice

- Learn/practice **ROS2 Jazzy** (Robot Operating System)
- We use Ubuntu 24.04
- Implementation and comparison of state-of-robotic perception algorithms in practice
- Real publicly available dataset
- Your own data! On-board sensors Real robot
- Map mini-city!

C++ refresher: https://youtu.be/F_vlB3yjxaM

- **Final goal:** prepare you to perform state-of-the-art research and implementation in robotics (not necessarily in robot perception)

ROS



ARCPro

Topics covered in CS4733/5733:

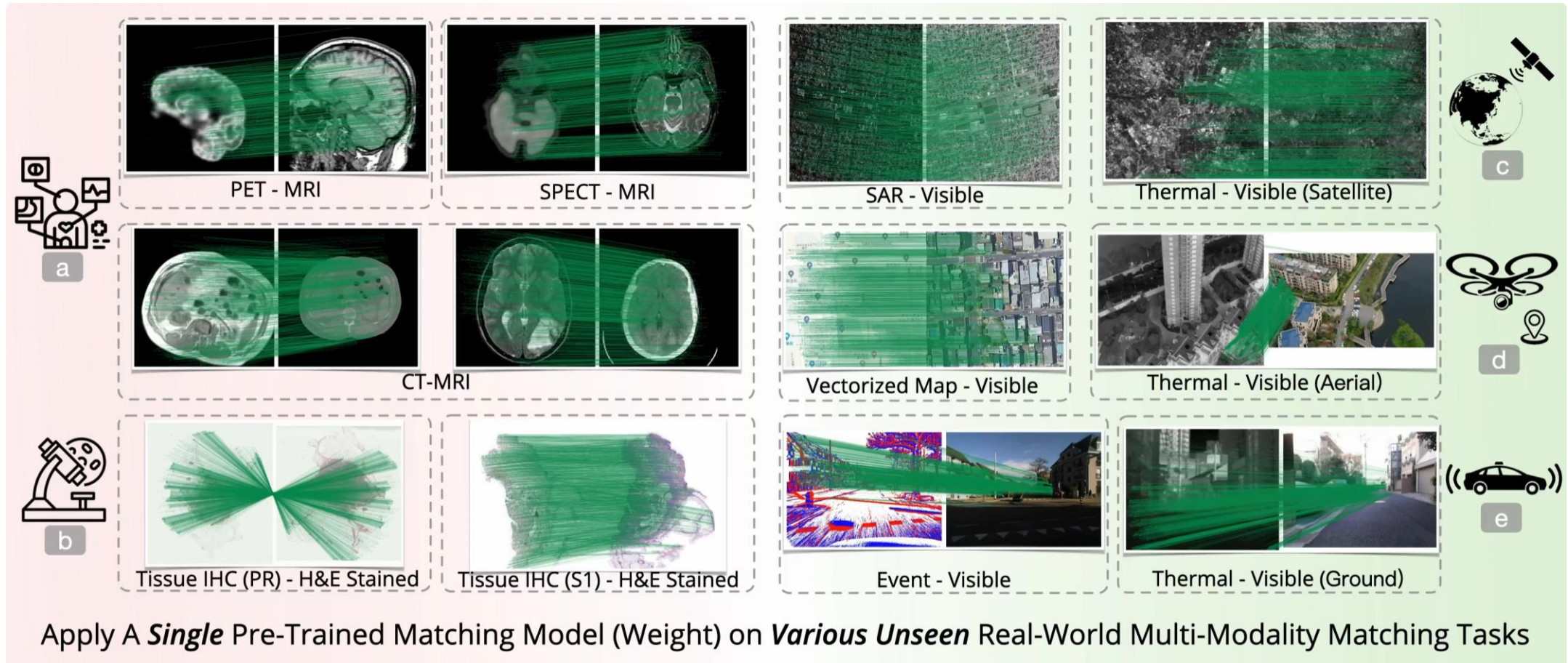
- 3D Geometry, transformation
- 2D multi-view geometric vision
- Visual Inertial Odometry and state estimation
- Place Recognition
- Visual Simultaneous Localization And Mapping (Visual SLAM)
- 3D object detection and tracking

Topics covered in CS4733/5733, continued

- LLM-based SLAM
- Beyond vision: multi-sensors fusion (camera, LiDARs and Radars)
- 3D trajectory and motion planning and optimization
- Multi-agent communication and decentralized algorithms
- Multi-agent coordination and navigation
- Collaborative SLAM
- Robustness of Visual navigation

LLM and Feature Matching!

Universal Cross-Modality Image Matching with Large-Scale Pre-Training



The course online Resources?

- Inspired by VNAV course at MIT:
 - Open Source: <https://ocw.mit.edu/courses/16-485-visual-navigation-for-autonomous-vehicles-vnav-fall-2020/>
 - Materials are adapted from this course

Course Schedule and Logistic

- **Lectures:** Tuesdays and Thursdays, 1:30-2:45 PM CEC 438
- **Office Hours/Lab sessions :**
 - When: Tu/Th, 12-13
 - Where: Rawl Engineering Practice Facility (REPF) – Room B4

- **Robotics experience:**
 - REPF B4



Prerequisites

- CS2413 or CS2414 or CS5005 and MATH 3333 or instructor's permission
- **Good to have knowledge in:**
 - Optimization Theory and calculus
 - Control Theory
 - Computer Vision
 - 3D Geometry
- **Coding skills:**
 - Familiarity with coding and experience in C++
 - Projects are largely based on ROS and OpenCV and all in C++.
- Previous experience in robotics and programming in ROS is a plus.

Required Software and Hardware

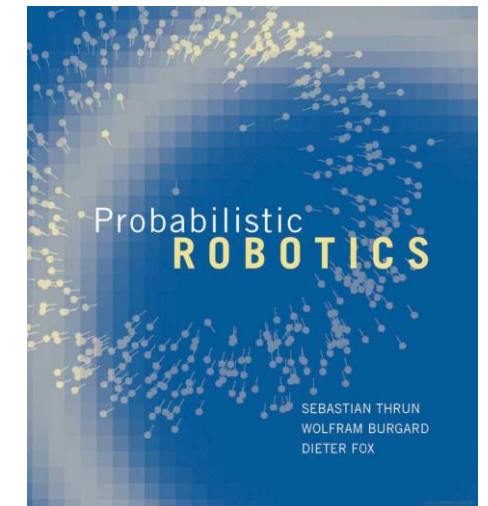
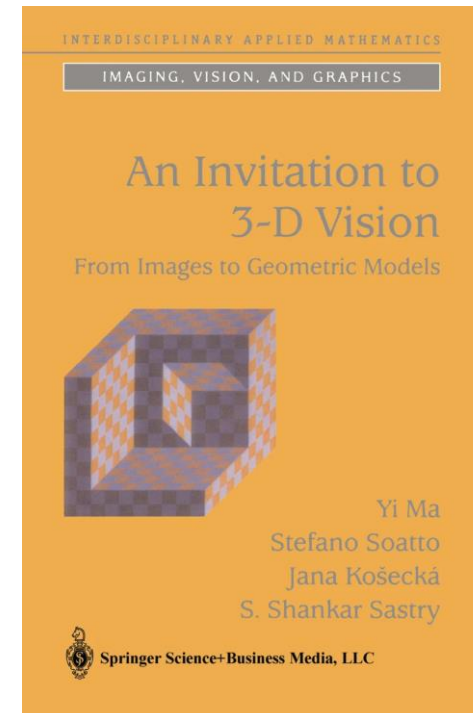
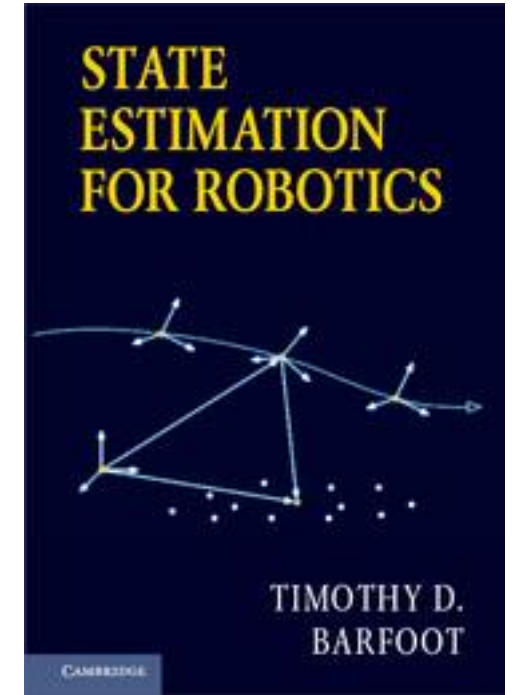
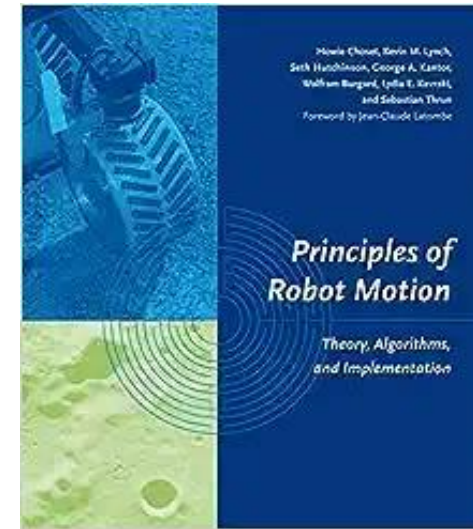
- **Ubuntu 24.04 :**
 - Dual boot, single boot
 - We cannot support other operating systems and students are responsible to resolve their issue if they use other OS
 - Please see the instruction in Lab 1
 - VM is available, but it is not recommended outside of campus
 - We may provide some computers with Ubuntu 24.04 installed in REPF B4
 - All students should get access to REPF, B4 by tomorrow. Please check!
- **ROS2Jazzy:** most of the lab assignments are in ROS2 Jazzy
- **Race car for some labs and final project:** will be distributed next week

Some notes on this course:

- This course is about one of the important part of **robotics** (i.e., mapping and perception), if you do not like robotics, you would not like this course
- This course includes a lot of **math**, concepts in geometry, optimization, linear algebra. If you do not like math, you would not like this course
- We use machine learning for some parts of this course, but this course is **not a ML course**.
- This course includes a lot of **programming** (mainly in C++). If you do not like programming at all (i.e., C++), you would not enjoy this course
- The final project and some labs includes **hands-on robotic experience**, if you do not like robotics and hardware experiments, you would not enjoy the course and projects.

Recommended Textbooks

- Books are recommended, but not required
- Some of them are available online:
 - [An invitation to 3-D vision](#)
 - [State Estimation for Robotics](#)



Course Evaluation (tentative)

Course Evaluation Metrics (Section 4733)

Lab assignments	70%
Final Exam	20%
Attendance and quizzes	5%
Teammate assessment and class activity	5%

Course Evaluation Metrics (Section 5733)

Lab assignment	60%
Final Project	20%
Final Exam	10%
Attendance and quizzes	5%
Teammate assessment and class activity	5%

Lab assignments components & submission policy

- Theory section (individual)
- Programming section (in group of 3)
- All parts of Lab 1 and Lab 2 needs to be done individually
- **Lab submission:** Github, your individual and group repository will be cloned at midnight of the deadline
- The answers should be typed
- **Lab grading policy:**
 - 15% penalty for each day late after deadline
 - Up to five grace days for programming section only
 - Can be used for lab 3 and higher only
 - Maximum 2 grace days for each lab assignment (programming section only), then the penalty is applied)
 - Assume your raw grade is G_1 , your final grade if you submit d days after the deadline will be:
$$\text{Grade} = \max(0, G_1 (1 - 0.15d))$$

Final project (section CS5733)

- Final project:
 - Work in groups of 3
 - Goal: Mapping mini-city using visual Inertial Odometry and Visual SLAM
 - You have to use onboard Sensing
 - You can choose any state-of-the-art or your own Visual SLAM algorithm
 - Kimera, ORBSLAM, Hydra, etc
 - Bonus section: autonomous navigation
- Final Project Delivery:
 - Final demo: showcasing the project outcome
 - Team presentation
 - Technical report
- Final project is optional for CS4733 (extra 10%!)



ARCPro



OU mini city

Final Project Presentation

- School of Computer Science End of the semester Poster Presentation
- Great opportunity to share your work with your friends and colleagues.

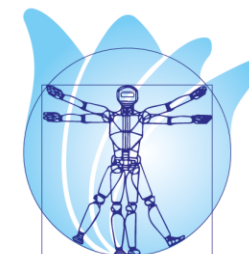


Final project as a research paper!

RSS: Early February
IROS: Early March
ICRA: September 15th
CVPR: November 15th



The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025
Tu June 10th - Tu June 17th, 2025
at the Music City Center, Nashville TN



**ROBOTICS
SCIENCE AND SYSTEMS**

JULY 15-19, 2024 | DELFT UNIVERSITY OF TECHNOLOGY, NETHERLANDS

Assignments for this week

- Background survey (participation will be graded):
 - When is due date? Thursday 8/28, 11:59 PM, will be closed after the deadline.
- Lab 1 (will be posted today!)
 - What is about? Installing Git, Ubuntu 24.04 and C++ basics
 - When is Due? Sept 2nd

Next Lecture

- Introduction to 3D Geometry
- Introduction to ROS2 (lab2)